

ET 8306 OPTICAL COMMUNICATION SYSTEM

CLASS ASSIGNMENT 2

28/01/2025

To be submitted on Friday 31/01/2025

QUESTION ONE

Draw the optical power loss model diagram for a point-to-point link and explain the concept of link power budget.

QUESTION TWO

It is required to construct an optical fiber communications link 5 km long with an analog modulation bandwidth of 10 MHz. The proposed system uses multimode step index fiber with core and cladding refractive indices of 1.48 and 1.46 respectively. At the operating wavelength the fiber exhibits an attenuation of 3db/km. The emitter is an LED which can launch 200 μ W of optical power into the fiber and the detector is a p-I-n photodiode which requires -40 dBm to achieve the required signal-to-noise ratio (SNR). One splice with 0.5dB loss is required and the connector at the detector has a loss of 1dB. The detector and emitter have rise times of 100 ns and 10 ns respectively. Calculate:-

- i. Coupling efficiency
- ii. Total power budget
- iii. Total system Loss
- iv. Power margin
- v. System rise time
- vi. Determine whether the link should operate satisfactorily and if not discuss the component changes you would recommend.

END